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15.6 Using the NLS panel data on N = 716 young women, we consider only years 1987 and 1988. We are interested in the relationship between $\ln(WAGE)$ and experience, its square, and indicator variables for living in the south and union membership. Some estimation results are in Table 15.10.

TABLE 15.10 Estimation Results for Exercise 15.6

	(1) OLS 1987	(2) OLS 1988	(3) FE	(4) FE Robust	(5) RE
C	0.9348	0.8993	1.5468	1.5468	1.1497
	(0.2010)	(0.2407)	(0.2522)	(0.2688)	(0.1597)
EXPER	0.1270	0.1265	0.0575	0.0575	0.0986
	(0.0295)	(0.0323)	(0.0330)	(0.0328)	(0.0220)
$EXPER^2$	−0.0033	−0.0031	−0.0012	−0.0012	−0.0023
	(0.0011)	(0.0011)	(0.0011)	(0.0011)	(0.0007)
SOUTH	−0.2128	−0.2384	−0.3261	−0.3261	−0.2326
	(0.0338)	(0.0344)	(0.1258)	(0.2495)	(0.0317)
UNION	0.1445	0.1102	0.0822	0.0822	0.1027
	(0.0382)	(0.0387)	(0.0312)	(0.0367)	(0.0245)
N	716	716	1432	1432	1432

(standard errors in parentheses)

a. The OLS estimates of the $\ln(WAGE)$ model for each of the years 1987 and 1988 are reported in columns (1) and (2). How do the results compare? For these individual year estimations, what are you assuming about the regression parameter values across individuals (heterogeneity)?

The OLS estimates for 1987 and 1988 are very similar. The coefficient on $EXPER$ is positive in both years, 0.1270 in 1987 and 0.1265 in 1988, suggesting that more work experience is associated with higher wages. The coefficient on $EXPER^2$ is negative in both years, -0.0033 in 1987 and -0.0031 in 1988, indicating diminishing returns to experience.

The coefficient on $SOUTH$ is negative in both years, -0.2128 in 1987 and -0.2384 in 1988. This suggests that women living in the South tend to have lower wages than those not living in the South. The coefficient on $UNION$ is positive in both years, 0.1445 in 1987 and 0.1102 in 1988, implying that union membership is associated with higher wages, although the estimated effect is somewhat smaller in 1988.

Overall, the signs and magnitudes of the coefficients are quite stable across the two years. This suggests that the relationship between $\ln(WAGE)$ and the explanatory variables did not change much from 1987 to 1988.

For the individual year OLS estimations, we assume that the regression parameter values are the same across individuals within each year. In other words, all individuals share the same intercept and slope coefficients. Any individual heterogeneity, such as differences in ability, education, motivation, or other unobserved characteristics, is included in the error term and is assumed to be uncorrelated with the explanatory variables.

b. The $\ln(WAGE)$ equation specified as a panel data regression model is

$$\ln(WAGE_{it}) = \beta_1 + \beta_2 EXPER_{it} + \beta_3 EXPER_{it}^2 + \beta_4 SOUTH_{it} + \beta_5 UNION_{it} + (u_i + e_{it}) \quad \text{(XR15.6)}$$

Explain any differences in assumptions between this model and the models in part (a).

The panel data model is specified as

$$\ln(WAGE_{it}) = \beta_1 + \beta_2 EXPER_{it} + \beta_3 EXPER_{it}^2 + \beta_4 SOUTH_{it} + \beta_5 UNION_{it} + (u_i + e_{it}) \qquad (1)$$

Compared with the individual-year OLS regressions in part (a), this model combines the observations from 1987 and 1988 and treats them as panel data. Therefore, it assumes that the slope coefficients are the same across the two years. In other words, the effects of $EXPER$, $EXPER^2$, $SOUTH$, and $UNION$ on $\ln(WAGE)$ are assumed to be stable over time.

The main difference is that the panel model explicitly includes the individual-specific effect u_i. This term captures unobserved characteristics of each individual that do not change over time, such as ability, education background, motivation, or other time-invariant personal traits. Thus, the error term is decomposed into two parts: u_i, the individual-specific component, and e_{it}, the idiosyncratic error term.

In this model, individuals are allowed to have different intercepts through u_i, since the effective intercept for individual i is $\beta_1 + u_i$. However, the slope coefficients are still assumed to be the same for all individuals.

If the model is estimated using fixed effects, u_i is allowed to be correlated with the explanatory variables. If it is estimated using random effects, u_i is assumed to be uncorrelated with all explanatory variables. Therefore, the random effects assumption is stronger, while the fixed effects approach is more flexible in controlling for individual heterogeneity.

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c. Column (3) contains the estimated fixed effects model specified in part (b). Compare these estimates with the OLS estimates. Which coefficients, apart from the intercepts, show the most difference?

Column (3) reports the fixed effects estimates. Compared with the OLS estimates in columns (1) and (2), the fixed effects estimates are noticeably different for some variables.

The coefficient on $EXPER$ falls from about 0.127 in the OLS estimates to 0.0575 in the fixed effects model. This suggests that once individual fixed effects are controlled for, the estimated effect of experience on wages becomes much smaller.

The coefficient on $EXPER^2$ also becomes smaller in absolute value. It changes from about -0.0032 in the OLS estimates to -0.0012 in the fixed effects estimate. This means that the diminishing return to experience is weaker under fixed effects.

The coefficient on $SOUTH$ becomes more negative. It is -0.2128 in 1987 and -0.2384 in 1988 under OLS, but it becomes -0.3261 under fixed effects. This suggests that the negative wage effect of living in the South is larger after controlling for individual fixed effects.

The coefficient on $UNION$ decreases from 0.1445 in 1987 and 0.1102 in 1988 to 0.0822 under fixed effects. Thus, the estimated positive effect of union membership becomes smaller after controlling for individual heterogeneity.

Apart from the intercepts, the coefficients that show the largest differences are $SOUTH$ and $EXPER$. If we choose one, $SOUTH$ shows the most noticeable difference in absolute terms, since its coefficient becomes much more negative in the fixed effects model. The coefficient on $EXPER$ is also substantially different because it falls by more than half compared with the OLS estimates.

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d. The F-statistic for the null hypothesis that there are no individual differences, equation (15.20), is 11.68. What are the degrees of freedom of the F-distribution if the null hypothesis (15.19) is true? What is the 1% level of significance critical value for the test? What do you conclude about the null hypothesis.

$$F = \frac{(SSE_R - SSE_U)/(N-1)}{SSE_U/(NT - N - K_S)} \tag{15.20}$$

$$H_0 : \beta_{11} = \beta_{12}, \beta_{12} = \beta_{13}, \ldots, \beta_{1,N-1} = \beta_{1N} H_1 : \text{the} \beta_{1i} \text{are not all equal} \tag{15.19}$$

The null hypothesis is that there are no individual differences, meaning that all individual-specific effects are zero:

$$H_0 : u_1 = u_2 = \cdots = u_N = 0. \tag{2}$$

The alternative hypothesis is that at least one individual effect is different from zero. There are $N = 716$ individuals and $T = 2$ years, so the total number of observations is $NT = 1432$. The numerator degrees of freedom are

$$df_1 = N - 1 = 715. \tag{3}$$

There are four slope regressors, $EXPER$, $EXPER^2$, $SOUTH$, and $UNION$, so the denominator degrees of freedom are

$$df_2 = NT - N - K = 1432 - 716 - 4 = 712. \tag{4}$$

Therefore, the relevant distribution is approximately

$$F_{715,712}. \tag{5}$$

At the 1% significance level, the critical value is about

$$1.19. \tag{6}$$

Since the reported test statistic is

$$F = 1.68, \tag{7}$$

and

$$1.68 > 1.19, \tag{8}$$

we reject the null hypothesis. Therefore, there is strong evidence of significant individual differences. This means that individual effects are important, and a pooled OLS model that ignores individual heterogeneity is likely inappropriate. A fixed effects model is more suitable because it controls for time-invariant individual characteristics.

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e. Column (4) contains the fixed effects estimates with cluster-robust standard errors. In the context of this sample, explain the different assumptions you are making when you estimate with and without cluster-robust standard errors. Compare the standard errors with those in column (3). Which ones are substantially different? Are the robust ones larger or smaller?

Column (4) reports the fixed effects estimates with cluster-robust standard errors. The coefficient estimates are exactly the same as those in column (3). The only difference is the way the standard errors are computed.

Without cluster-robust standard errors, as in column (3), we are assuming that the idiosyncratic errors are homoskedastic and uncorrelated across time for the same individual. In other words, the usual fixed effects standard errors assume that the error terms are relatively well behaved.

With cluster-robust standard errors, we allow the error terms to be correlated within each individual over time. Since this sample contains two years, 1987 and 1988, clustering by individual allows the error term for the same woman in 1987 to be correlated with her error term in 1988. It also allows for heteroskedasticity across individuals. However, we still assume independence across different individuals.

Comparing the standard errors in columns (3) and (4), most of them become larger when cluster-robust standard errors are used. The standard error of the intercept increases from 0.2522 to 0.2688. The standard error of $UNION$ increases from 0.0312 to 0.0367. The most substantial difference is for $SOUTH$, whose standard error increases from 0.1258 to 0.2495, almost doubling.

The standard error of $EXPER$ is almost unchanged and is slightly smaller, falling from 0.0330 to 0.0328. The standard error of $EXPER^2$ remains essentially the same at 0.0011.

Therefore, the main difference is for $SOUTH$. Under the usual FE standard errors, the t-statistic for $SOUTH$ is approximately

$$\frac{-0.3261}{0.1258} \approx -2.59. \tag{9}$$

Using the cluster-robust standard error, it becomes

$$\frac{-0.3261}{0.2495} \approx -1.31. \tag{10}$$

Thus, $SOUTH$ appears significant under the usual fixed effects standard errors, but it is no longer significant when cluster-robust standard errors are used. This suggests that ignoring within-individual correlation or heteroskedasticity may understate the uncertainty of the estimate.